

Performance Analysis of Grover's Search Algorithm Under Realistic Noise Models on NISQ Simulators

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Abstract: Grover's search algorithm provides a provable quadratic speedup over classical unstructured search, making it one of the most relevant quantum algorithms for near-term deployment. However, current quantum hardware operates in the Noisy Intermediate-Scale Quantum (NISQ) regime, where decoherence and gate errors severely degrade algorithm performance. This research presents a simulation-based comparative study of Grover's algorithm under three realistic noise models: depolarizing noise, bit-flip errors, and phase-flip errors. Using Python and the Qiskit Aer simulator, Grover's search circuit is implemented for search spaces of $n = 2, 3, 4, 5$ qubits. Noise is injected at parameterized error rates swept across the range $p \in [0.001, 0.1]$ and success probability is measured at each configuration. The primary objective is to identify the critical noise threshold p^* at which Grover's algorithm loses its quantum advantage over classical brute-force search, and to determine how this threshold scales with the number of qubits. Results include empirical noise threshold curves for each noise type, crossover analysis comparing Grover's performance against the classical baseline $\frac{1}{N}$, and a reproducible Qiskit codebase. This work directly addresses a gap in the existing literature by systematically identifying quantum advantage loss thresholds across multiple noise types using modern quantum simulation tools.

Keywords: Grover's algorithm, NISQ computing, noise models, quantum advantage

I. Introduction

Quantum computing leverages the principles of superposition, entanglement, and quantum interference to solve certain computational problems significantly faster than classical computers. Among the most practically relevant quantum algorithms is Grover's search algorithm [1], which achieves $O(\sqrt{N})$ query complexity for searching an unstructured database of N entries, compared to the classical $O(N)$ average case. This quadratic speedup has wide applications in optimization, cryptanalysis, and database search, making Grover's algorithm a standard benchmark for evaluating near-term quantum hardware.

Despite its theoretical promise, quantum algorithms face a fundamental challenge in practice: noise. Physical qubits are

susceptible to environmental decoherence, gate imperfections, and measurement errors. These noise sources collectively corrupt the quantum interference patterns that Grover's algorithm depends upon to amplify the probability of measuring the correct answer. Present-day quantum processors operate in the Noisy Intermediate-Scale Quantum (NISQ) regime, characterized by gate error rates on the order of 10^{-3} to 10^{-2} and coherence times in the microsecond to millisecond range. Under these conditions, even moderately deep circuits such as Grover's can suffer severe degradation in success probability.

While previous research has studied noise effects on Grover's algorithm, no work has systematically combined modern Qiskit-based circuit simulation with explicit identification of the quantum advantage loss threshold p^* across multiple isolated noise channels. This paper analyzes a systematic empirical study of Grover's algorithm under realistic noise conditions using the Qiskit Aer simulator. By sweeping noise error rates across a controlled range and measuring success probability for each configuration, this work aims to identify the precise noise threshold beyond which Grover's algorithm loses its quantum advantage over classical search.

A. Contributions

This work extends the current literature by making the following specific contributions:

- 1) Implementation of a comprehensive Qiskit-based simulation framework for Grover's algorithm across $n = 2, 3, 4, 5$ qubit configurations with systematic noise injection
- 2) Empirical identification of the quantum advantage threshold p^* for three distinct noise models through controlled error rate sweeps from $p = 0.001$ to $p = 0.1$
- 3) Explicit crossover analysis comparing Grover's noisy performance against the classical random search baseline $\frac{1}{2^n}$
- 4) Development of a reproducible open-source Qiskit codebase suitable for extension to noise mitigation and error correction research

The remainder of this paper is organized as follows: Section II reviews related work and identifies gaps in existing literature. Section III concludes with a summary of expected outcomes and future research directions.

II. Literature Review

This section reviews prior work on noise effects in Grover’s algorithm and identifies the specific research gap this work addresses.

A. Ijaz and Faryad (2023) – Base Paper

Ijaz and Faryad [2] conducted a comprehensive noise analysis of Grover’s algorithm using Qiskit Aer for $n = 3, 4, 5$ qubits with 10,000 shots per configuration. Four noise channels were studied: depolarizing, bit-flip (X), phase-flip (Z), and bit-phase flip (Y). The key finding was that success probability decays exponentially with error probability p but only linearly with qubit count n . The bit-phase flip channel caused the most severe degradation across all configurations.

Strength: Real Qiskit Aer simulation with four noise types providing empirical confirmation of exponential decay relationship between success probability and error rate.

Limitation: Does not identify the critical threshold p^* where quantum advantage over classical search is lost. The study focuses on comparing standard versus QSVT implementations rather than explicit threshold analysis.

B. Salas (2008)

Salas [3] performed the earliest systematic numerical study of noise in Grover’s algorithm using Fortran-based Monte Carlo simulation for $n = 2$ to 7 qubits under a depolarizing channel model. The study derived an exponential damping law for success probability and an allowed-error law: $\varepsilon_{th}(N) \sim N^{-1.1}$, where $N = 2^n$. An absolute threshold of $\varepsilon > 0.043$ was identified above which the algorithm fails for any qubit count.

Strength: First quantitative noise threshold derived for Grover’s algorithm providing foundational analytical results that remain cited in modern literature.

Limitation: Uses an outdated Fortran framework predating Qiskit. Studies only depolarizing noise without explicit comparison against classical search advantage boundary.

C. Dowarah et al. (2025)

Dowarah et al. [4] applied Random Matrix Theory (RMT) to model the noisy Grover operator as a Floquet unitary and derived two analytical noise thresholds. The computing transition at $\delta_{c,comp} \sim O(2^{-L/2})$ marks where quantum advantage is lost, and the ergodicity transition at $\delta_{c,gap} \sim O(L^{-1})$ marks where circuit structure collapses entirely.

Strength: State-of-the-art analytical framework identifying two distinct thresholds with exact scaling laws based on rigorous theoretical foundations.

Limitation: Entirely theoretical work restricted to systematic coherent noise. No Qiskit simulation or empirical validation provided to confirm predictions on actual quantum circuits.

D. Srivastava et al. (2025)

Srivastava et al. [5] proposed using quantum switches exploiting indefinite causal order to mitigate depolarizing noise in Grover’s algorithm. Two frameworks were developed,

with the second framework (delayed measurement) providing significantly greater noise resilience by preserving correlations between the input state and quantum switch register throughout all iterations.

Strength: Novel mitigation approach demonstrating improved success probability analytically. Framework is directly applicable to NISQ devices for practical noise reduction.

Limitation: Entirely analytical with no Qiskit simulation to validate performance improvements. Studies only depolarizing noise and does not identify the threshold p^* for standard Grover circuits without mitigation.

E. Comparative Analysis and Research Gap

Table I summarizes the strengths and limitations of reviewed literature. The comparative analysis reveals that no existing study combines (1) a modern Qiskit simulation framework, (2) multiple isolated noise types, (3) systematic error rate sweeps, and (4) explicit identification of the quantum advantage loss threshold.

TABLE I
COMPARATIVE ANALYSIS OF RELATED WORK

Paper	Method	Strength	Limitation
Ijaz & Faryad (2023) [2]	Qiskit Aer simulation	4 noise types; exponential decay confirmed	No p^* threshold; QSVT focus
Salas (2008) [3]	Monte Carlo, Fortran	First threshold law derived	Outdated tools; depolarizing only
Dowarah et al. (2025) [4]	Random Matrix Theory	Two-threshold analytical model	No simulation; coherent noise only
Srivastava et al. (2025) [5]	Analytical, quantum switch	ICO mitigation; improved success	No Qiskit; depolarizing only

This work fills the identified gap by implementing a systematic Qiskit-based empirical study that explicitly locates the quantum advantage threshold p^* across three noise types through controlled error sweeps, providing concrete benchmarks for NISQ hardware calibration.

III. Conclusion

This research addresses a critical gap in quantum computing literature by systematically identifying the noise threshold at which Grover’s search algorithm loses its quantum computational advantage over classical brute-force search. By combining modern Qiskit-based simulation with comprehensive noise sweeps across depolarizing, bit-flip, and phase-flip channels, this work will deliver empirical threshold curves that are directly actionable for NISQ hardware development and calibration strategies.

The expected outcomes include: (1) empirical noise threshold values p^* for each noise type and qubit configuration, (2) scaling analysis showing how thresholds evolve with increasing system size, (3) crossover plots explicitly marking the quantum advantage boundary, and (4) a reproducible open-source Qiskit codebase suitable for extension to error mitigation research.

Compared to Ijaz and Faryad (2023), this work will produce the explicit quantum advantage threshold map their study does not provide. Compared to Salas (2008), results will be validated on modern Qiskit circuit simulators with three noise types rather than one. The threshold values obtained will serve as direct empirical inputs for noise mitigation strategies such as the quantum switch framework of Srivastava et al. (2025), and will provide the circuit-level validation that the analytical RMT framework of Dowarah et al. (2025) presupposes.

This work establishes a foundation for future research in quantum algorithm robustness under realistic noise conditions, providing concrete benchmarks for near-term quantum hardware capabilities and limitations.

References

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